

Department of EEE, AUST
EEE 3218 (Digital Signal Processing Lab)
Fall 2025

*****Please select and submit your paper title **by October 3** and obtain approval from your course teacher **by October 5**.

When submitting the paper title, make sure to include the **student ID of every group member**.

Please note that **no two groups can select the same paper**. If another group has already selected and submitted your chosen paper in the Excel sheet, you must select a different paper.

Please check the Excel sheet before submitting your paper to ensure that your selected paper has not already been taken by another group.*****

Submission Link: https://docs.google.com/spreadsheets/d/1mErhx5M_EvEVA7ckMlfhyLAR7ni4YBG6vZ2uuSIg8o/edit?usp=sharing

Title:

Reproduction and Critical Analysis of a Digital Signal Processing (DSP) Research

Study Objectives & Project Guidelines:

This project is designed to help students develop skills in scientific research, implementation, and critical evaluation of DSP techniques. The project will be conducted in structured phases, ensuring both technical and critical thinking development.

Phase 1: Research Paper Selection and Approval

1. Each student or group (consisting of a maximum of four members) must find a research article related to DSP topics (e.g., time-frequency analysis, adaptive filters, speech processing, etc.).
2. Recommended platforms for finding research articles:

- Google Scholar
- IEEE Xplore
- ResearchGate
- ScienceDirect
- SpringerLink

3. Submit the selected paper (PDF) along with a short summary (max 1 page) addressing:
 - The problem being addressed
 - The methods used
 - The specific segment of the paper they aim to reproduce
4. The instructor will review and approve the paper based on relevance and feasibility.
5. Only after approval can students proceed to the next phase.

Phase 2: Understanding the Research Work

1. Thoroughly study the approved paper, focusing on:
 - DSP concepts used
 - Mathematical models and algorithms
 - Datasets and input signals
2. Clarify any assumptions and identify the exact portion of the work that will be reproduced.

Phase 3: Implementation and Reproduction

- ## Phase 4: Analysis and Validation

- ## Phase 5: Report Writing and Critical Evaluation

2. Introduction: Outline the problem, importance of the topic, and relevance to DSP.
3. Reproduced Segment: Describe the portion of the paper reproduced, algorithms, and steps taken.
4. Results and Analysis: Present results with comparisons, graphs, and discussion on deviations and limitations.
5. Critical Evaluation: Discuss what makes the method unique, what challenges they faced, and how results could be improved.
6. Software/Implementation Details: Mention what tools, versions, and code snippets were used.
7. Conclusion: Summarize findings and reflect on learning outcomes.

Activities	Time Frame
OEL Assignment Given	3 weeks before the assessment day
Students discuss among themselves on the task	Within the 3-week timeframe
Students simulate the experiment	Before the assessment
Students prepare the OEL report	Before the assessment
Students demonstrate the OEL experiment	On the assessment day (Date fixed by respective course teacher)
Student Submit the OEL report	During or before the assessment (Both Hardcopy and Softcopy)

Sl. No.	CO Statement	P Os	Bloom's Domain			K	P	A	Teaching Learning Strategy	Assessment Method
				C A P						

1	Investigate issues related to Digital Signal Processing to Design and conduct experiments and data analysis	4	4		2	K1-K4, K5, K6, K8	P1, P4, P7		Lab Demonstration	OEL Demonstration, OEL Report
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P's (Complex Engineering Problems) addressed through this open-ended lab.

P1: This OEL requires knowledge of

Digital signal processing and programming (K3)

Design Knowledge (K5)

Simulation and Testing (K6)

Previously conducted works in the same domain (K8)

P4: Familiarity of issues

The experiment is involved with infrequently encountered issues. To remove noise from audio signals, students must know about many things related to medical sciences.

P7: Interdependence

The design has several subsystems like sampling, filter design, frequency domain, and time domain analysis, etc.

Assessment Criteria and Rubrics

S N	Component s	Excellent	Good	Basic	Acceptabl e	Unacce pta ble	Assess m ent Method	Mar ks
1	Backgrou nd Knowled ge & Understa nd ing	6–5: Thorough underst an ding; all questions answered correctly	5–4: Good underst an ding; most answer ed correctly	4–3: Moder ate underst an ding	2: Limited underst an ding	1–0: Very poor	Demo / Viva	6
2	Paper Selection & Justificati o n	4: Highly relevant, clear scope, strong justifica tio n	3: Relevant with minor gaps	2: Somewh at relevant	1: Weak justifica tio n	0: Not approved	Propo sa l Subm iss ion	4

3	Implementation / Reproduction	6–5: Correct reproduction with proper execution	5–4: Mostly correct	4–3: Partial (≥50%)	2: Attempt but incorrect	1–0: No implementation	Code Demo	6
4	Results & Analysis	4: Accurate results + strong analysis	3: Good results with some analysis	2: Results but weak analysis	1: Results without analysis	0: Missing	Report + Viva	4
5	Report Writing (Total 10 Marks)	10–9: Excellent structure, detailed explanation, clear reproduction, proper figures, software	8–7: Good structure with minor issues	6–5: Adequate but lacks clarity in sections	4–3: Poor organization and weak explanation	2–0: Incomplete report	Report Check	10

		well documented						
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Appendix: A

Supportive Documents

1.What is Open Ended Lab (OEL)?

a. An open-ended laboratory is where students are given the freedom to develop their own experiments, provided with clear rubrics, autonomy, and opportunities to self-reflect on their performance, instead of merely following the already set guidelines from a lab manual or elsewhere. b. Creativity and innovation flourishes in place where open ideas are promoted c. No pre-planned lab manual is given by the instructor.

- d. No help from the faculty or lab demonstrator.
- e. Problems and constraints are given to address Knowledge, complex engineering Problem and Activities (KPA)s conforming to CO/PO/PEO.
- f. Objectives are to measure the **students' teamwork, report writing, and presentation (attitude/behavior test)**; demonstration of practical skills through the practical tests to measure the psychomotor ability.
- g. It will create the students to think critically and also out of the box.
- h. The students here have to devise their own strategies and back them with explanations, theory and logical justification.

The concept of different levels of openness was discussed by many researchers. The areas of concern are categorized as problems, ways and means, and answers. One such level of openness is shown in Table 1.

Table-1: Level of Openness in OEL					
Level	Problems	Ways & Means	Answers	Common Name	Degree of Openness
0	Given	Given	Given	Close	0%
1	Given	Given	Open	Partial Open	33%
2	Given	Open	Open	Partial Open	67%
3	Open	Open	Open	Fully Open	100%

2. In a fully open-ended laboratory activity- Students will be exposed to a practical problem to measure their skills Provides a suitable method to measure the students' psychomotor ability. They are required (as a group) to determine the objectives and scopes, identifying the required apparatus and preparing the methodology, running the experiment, presenting or demonstrating the work, and finally submitting the technical report.

3. First 5 weeks have regular lab session and then provide an OEL. The EOL should be Fully Open. Student will conduct an EOL based on the experiments they have done in the lab and knowledge earned from the theory course.

4. Table for different domains in Bloom's Taxonomy

Level	Cognitive	Affective	Psychomotor
1	Remember	Receiving	Imitation
2	Understand	Responding	Manipulation
3	Apply	Valuing	Precision
4	Analyze	Organization	Articulation
5	Evaluate	Characterization	Naturalization
6	Create		

- I. Imitation: Observing and copying basic actions or movements.
- II. Manipulation: Performing learned skills with some degree of proficiency and coordination.
- III. Precision: Demonstrating accurate and controlled performance of skills with refinement and consistency.
- IV. Articulation: Adapting and combining skills to create more complex and coordinated actions.
- V. Naturalization: Performing skills automatically and with a high level of proficiency, often without conscious effort.

5.

Depth of knowledge required: Cannot be resolved without in-depth engineering knowledge at the level of one or more of WK3, WK4, WK5, P1 WK6 or WK8 which allows a fundamentals-based, first principles analytical approach

P2 Range of conflicting requirements

P3 Depth of analysis required: Have no obvious solution and require abstract thinking, originality in analysis to formulate suitable models

P 4	Familiarity of issues
P 5	Extent of applicable codes
P 6	Extent of stakeholder involvement and level of conflicting requirements
P 7	Interdependence: Are high level problems including many component parts or sub-problems